

Soccer TIPS Engineering Department

Document: Layer 1_v0.1 General Developer Handoff Notes

General Developer

Layer 1 v0.1 Implementation Handoff Notes

Updated after Lead Developer review to include current status, run commands, future-helper usage guidance, and corrected subtitle.

Current Layer 1 v0.1 Status

- Layer 1 v0.1 supports provider-specific loading for SkillCorner and Metrica.
- It produces canonical player and ball tracking outputs.
- It writes validation reports.
- It supports controlled batch processing.
- It supports QA inspection and schema compliance checks.
- It does not include orientation normalization, tactical metrics, Layer 2 feature engineering, dashboards, AI interpretation, or reporting.

Future Code Helper Usage Note

Future code helpers should use this document together with:

- Engineering Department Doc, Validation & Data Ingestion Handoff Summary
- AIB Project System Architecture / Workflow Specification
- This document explains what was built.
- The validation handoff explains why the ingestion rules exist.
- The architecture document explains how Layer 1 connects to the full Soccer TIPS system.

How to Run Layer 1 v0.1

Open the terminal from the project root folder. On the current Windows setup, that project root is:

C:\Users\amsch\. Personal Work\Projects\Soccer TIPS

In VS Code, this means opening the Soccer TIPS project folder and using the integrated PowerShell terminal from that root folder. Commands below assume PowerShell on Windows.

Item	Current value / command
Activate virtual environment, if applicable	.\venv\Scripts\Activate
Install requirements	pip install -r requirements.txt
Set PYTHONPATH for src-layout imports	\$env:PYTHONPATH="src"

Item	Current value / command
Run sanity tests	python tests/test_layer1_ingestion.py
Process one SkillCorner match	python -m soccer_tips.data.build_layer1 --provider skillcorner --skillcorner-match-id 2017461
Process one Metrica sample game	python -m soccer_tips.data.build_layer1 --provider metrica --metrica-game Sample_Game_1
Run full controlled batch mode	python -m soccer_tips.data.build_layer1 --provider all --batch
Run Layer 1 QA inspection utility	python -m soccer_tips.data.qa_layer1_outputs

Recommended Run Order

1. Open PowerShell from the Soccer TIPS project root.
2. Activate the virtual environment if the project uses one.
3. Install requirements.
4. Set PYTHONPATH to src.
5. Run the sanity tests.
6. Run a single SkillCorner match.
7. Run a single Metrica sample game.
8. Run full controlled batch mode.
9. Run the Layer 1 QA inspection utility.

```

.\env\Scripts\Activate
pip install -r requirements.txt
$env:PYTHONPATH="src"
python tests/test_layer1_ingestion.py
python -m soccer_tips.data.build_layer1 --provider skillcorner --skillcorner-match-id 2017461
python -m soccer_tips.data.build_layer1 --provider metrica --metrica-game Sample_Game_1
python -m soccer_tips.data.build_layer1 --provider all --batch
python -m soccer_tips.data.qa_layer1_outputs

```

Output Locations

Item	Current value / command
SkillCorner processed outputs	data/processed/skillcorner/{match_id}/
Metrica processed outputs	data/processed/metrica/{sample_game}/
Validation reports	data/processed/validation_reports/
Layer 1 QA summary JSON	data/processed/validation_reports/layer1_v01_qa_summary.json

Expected Processed Output Files

- SkillCorner runs write canonical_tracking_players.csv, canonical_tracking_ball.csv, match_metadata.json, players_metadata.csv, phases_of_play.csv, and dynamic_events.csv when available.
- SkillCorner match 1953632 does not write dynamic_events.csv because that optional source file is missing in the raw dataset.
- Metrica Sample_Game_1 and Sample_Game_2 write canonical_tracking_players.csv, canonical_tracking_ball.csv, and events.csv.
- Metrica Sample_Game_3 is intentionally out of scope for Layer 1 v0.1 batch processing.

Acceptable Expected Warnings

- SkillCorner match 1953632 may report pass_with_warnings because its optional dynamic_events.csv file is missing.
- Metrica Sample_Game_1 may report pass_with_warnings because it contains one known kickoff initialization event with Start Frame = 1 and End Frame = 0.
- These warnings are acceptable because they match validated dataset behavior and do not prevent canonical tracking outputs from being written.

Warnings or Errors That Are Not Acceptable

- Any validation status of error or fail.
- Missing canonical_tracking_players.csv or canonical_tracking_ball.csv for a processed run.
- Schema compliance warnings for canonical_tracking_players.csv or canonical_tracking_ball.csv.
- Missing required SkillCorner metadata, tracking, or phases files.
- Missing required Metrica home tracking, away tracking, or events files for Sample_Game_1 or Sample_Game_2.
- Frame continuity failures or duplicate frame failures that were not previously documented as expected provider behavior.
- Any new provider-specific issue that changes canonical schema expectations, coordinate handling, or future Layer 2 assumptions.

Chronological Development Notes

The sections below preserve the General Developer work history. Earlier limitations are historical and may be superseded by later development sections.

First Development

1. Files created or changed: Created the full v0.1 Layer 1 ingestion package. These files replaced initial files with the same names where applicable.
2. What each file does: config.py centralizes paths; schemas.py defines canonical outputs; common.py handles simple JSON/CSV/path utilities; normalize.py handles provider coordinate

conversion; skillcorner_loader.py handles SkillCorner raw loading and canonical conversion; metrica_loader.py handles Metrica raw loading and canonical conversion; validation.py creates structural validation reports; build_layer1.py runs provider processing and writes outputs; tests/test_layer1_ingestion.py runs lightweight sanity checks.

3. Key functions added: load_skillcorner_match(), load_metrice_game(), process_skillcorner_match(), process_metrice_game(), validate_skillcorner_loaded_match(), validate_metrice_loaded_game(), metrice_normalized_to_metric(), enforce_tracking_player_schema(), enforce_tracking_ball_schema().
4. Important assumptions: Raw files are under external/; processed outputs are reproducible and ignored by Git.
5. Tests performed: Ran python tests/test_layer1_ingestion.py; all included sanity checks passed.
6. Known limitations at this point: Needed real-data execution in the local repo. Orientation normalization, batch processing, packaging, and Layer 2 remained future work.
7. Lead Developer review: Not needed after every file, but recommended after the first real SkillCorner match, the first real Metrica sample game, and before moving to Layer 2 feature engineering.

Second Development

1. Files/output created: SkillCorner processed output for match 2017461; Metrica processed output for Sample_Game_1; validation reports for both providers.
2. Result: SkillCorner passed. Metrica passed with warnings.
3. Important interpretation: SkillCorner DtypeWarning was not a failure. Metrica warning was expected and needed confirmation in the JSON report.
4. Lead Developer review: Not urgent yet, but this was the first meaningful checkpoint. After confirming the Metrica warning and running Sample_Game_2, it would be reasonable to send the Lead Developer the validation output summary.

Third Development

Files/output created

- Processed SkillCorner outputs for 2017461.
- Processed SkillCorner outputs for 1953632.
- Processed Metrica outputs for Sample_Game_1.
- Processed Metrica outputs for Sample_Game_2.
- Validation reports for all processed runs.

What was confirmed

- SkillCorner standard match processing works.
- SkillCorner missing dynamic events handling works.
- Metrica Sample_Game_1 processes with expected known kickoff warning.
- Metrica Sample_Game_2 processes cleanly.
- Combined provider runner works.

Known warnings

- SkillCorner dynamic events CSV had mixed dtypes in some columns at this point.
- Metrica Sample_Game_1 has one invalid kickoff event range.
- SkillCorner 1953632 is missing optional dynamic events.

Known limitations at this point

- Still v0.1 only.
- No orientation normalization.
- No tactical metrics.
- No Layer 2 feature engineering.
- No batch-all-matches processing yet.

Lead Developer review

- This became a good checkpoint to send to the Lead Developer because the first real Layer 1 ingestion milestone was working.

Fourth Development

Files changed

- skillcorner_loader.py.
- build_layer1.py.
- tests/test_layer1_ingestion.py.

Files added

- qa_layer1_outputs.py.
- README_LAYER1_QA_CLEANUP.md.

What was cleaned

- SkillCorner dynamic events now use low_memory=False to suppress the mixed dtype warning.

What was added

- Controlled batch mode.
- Local QA output inspection.
- Schema compliance reporting.
- QA JSON summary output.

Schema compliance

- No canonical schema changes were made.
- The QA utility checks player and ball output columns against the existing canonical schemas.

Known warnings preserved

- SkillCorner 1953632 missing dynamic events remains a warning.
- Metrica Sample_Game_1 kickoff invalid event range remains a warning.

Lead Developer review

- No architecture-level schema change was made.
- After running batch mode and the QA utility, send the QA summary/output to the Lead Developer as the next checkpoint.

Fifth Development

Layer 1 v0.1 QA and cleanup pass is complete.

Files changed/added in this pass

- src/soccer_tips/data/skillcorner_loader.py.
- src/soccer_tips/data/build_layer1.py.
- src/soccer_tips/data/qa_layer1_outputs.py.
- tests/test_layer1_ingestion.py.

What was cleaned or added

- Cleaned SkillCorner dynamic events mixed dtype warning using low_memory=False.
- Added controlled batch mode.
- Added QA utility for processed output inspection.
- Added schema compliance checks against canonical schemas.
- Added structured QA summary output.

Batch processing result

- All 10 SkillCorner matches from matches.json processed.
- Metrica Sample_Game_1 processed.
- Metrica Sample_Game_2 processed.
- Metrica Sample_Game_3 remained out of scope.

Schema compliance

- All canonical_tracking_players.csv outputs passed schema checks.
- All canonical_tracking_ball.csv outputs passed schema checks.
- No schema changes were made.

Validation summary

- SkillCorner: 9 pass, 1 pass_with_warnings.
- Metrica: 1 pass, 1 pass_with_warnings.

Known warnings remaining

- SkillCorner 1953632 is missing optional dynamic events.
- Metrica Sample_Game_1 has one known invalid kickoff event range.
- Both are expected non-fatal warnings.

Lead Developer review

- Recommended now. This is a meaningful milestone: Layer 1 v0.1 now supports provider-specific loading, canonical outputs, validation reports, QA inspection, and controlled batch processing.
 - No Lead Developer decision is needed for schema changes because no schema changes were made.
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Scope Guard

- This document does not authorize Layer 2 tactical metrics.
- Do not add compactness, transition stability, tactical identity modeling, dashboards, AI interpretation, or reporting from this handoff alone.
- Before moving into Layer 2 feature engineering, use this document together with the validation handoff and architecture document, then obtain Lead Developer review.